

Questions by topic: Water treatment, conservation & new sources - Set 1 (65 marks)

Answer all questions in the spaces provided.

01 . 1

Describe **four** factors that should be considered when choosing a site for a dam and reservoir.

[4 marks]

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01 . 2

Give **one** advantage and **two** disadvantages of an estuary barrage compared with a reservoir on land.

[3 marks]

Advantage

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Disadvantage 1

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Disadvantage 2

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01 . 3

Explain why desalination of seawater is only used in some countries.

[2 marks]

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01 . 4

Explain how artificial recharge can make aquifer supplies sustainable.

[1 mark]

10

0 2

Table 1 shows the effect of installing water meters on the mean daily water use of households in a UK town.

Table 1

	Unmetered households (fixed charge)	Metered households
Mean water use / dm^3 per person per day	150	125
Percentage with low-flush toilets	35	58
Percentage using a garden hosepipe weekly in summer	44	20

0 2

. 1

Calculate the percentage reduction in water use per person in metered households.

[1 mark]

Answer

%

0 2

. 2

Explain why metering reduces water use.

[2 marks]

0 2

. 3

Describe **three** ways in which domestic water use can be reduced other than metering.

[3 marks]

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0 2 . 4 Explain why about 20% of the water intended for public supply in the UK never reaches consumers, and how this could be reduced.

[2 marks]

0 2 . 5 Explain how drip irrigation reduces the water needed to grow a crop compared with spray irrigation.

[2 marks]

10

0 3

Figure 1 lists the stages in a water treatment works supplying drinking water from a lowland river.

Figure 1

Stage	Process
1	Screens
2	Sedimentation
3	Aeration
4	Flocculation and clarification
5	Filtration through sand and gravel
6	Activated carbon
7	Sterilisation
8	pH adjustment

0 3**1**

Explain why clay particles do not settle out in the sedimentation stage, and describe how they are removed in stage 4.

[3 marks]

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0 3**2**

Explain the purpose of aeration (stage 3).

[2 marks]

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0 3**3**

Explain why ozone or UV light may be used for sterilisation instead of chlorine when water comes from peaty upland sources.

[2 marks]

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03 . 4

Describe how each of the following contaminants is removed from water.

[3 marks]

Toxic metal ions (lead, mercury, arsenic)

Salt

Odours and pesticides

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04 . 1

Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

- A Desalinated seawater
- B River water
- C Rainwater catchment
- D Newly discovered aquifers

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04 . 2

Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

04 . 3

Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

5

05 . 1

Which of the following is a feature of a river that affects its usefulness as a water source? Tick **one** box.

[1 mark]

- A Its total annual discharge
- B The albedo of its surface
- C Its salinity
- D The age of its bedrock

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05 . 2

Explain why rainwater collection is increasingly important in cities such as those in India.

[2 marks]

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05 . 3

Explain why an estuary barrage has fewer land-use conflicts than a reservoir on land but still causes environmental problems.

[2 marks]

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06

Table 1 shows the water quality required for four different uses.

06 . 1

Complete Table 1 by giving the quality requirement for each water use and the reason for it.

[4 marks]

Table 1

Water use	Quality requirement	Reason	
Potable (public) supply			ox
Spray irrigation			
Power station condenser cooling			
Industrial boiler water			

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06 . 2

Explain why aeration is used in the treatment of water from anaerobic sources.

[2 marks]

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06 . 3

Explain how flocculation removes clay particles that sedimentation cannot.

[2 marks]

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0 6 . 4 Give **two** methods used to sterilise treated water.

[2 marks]

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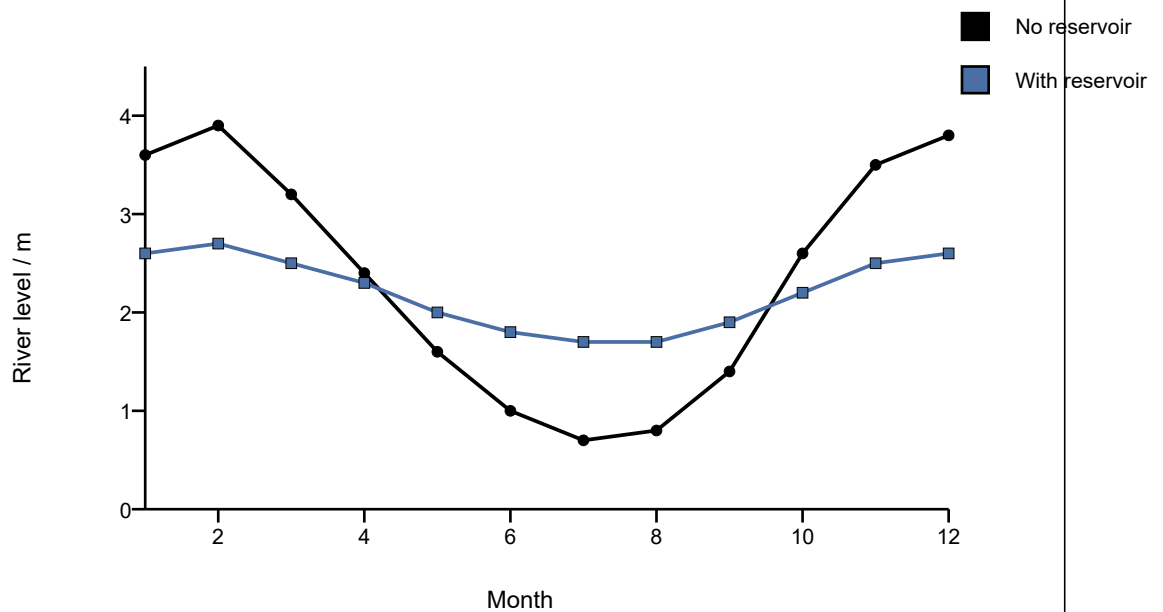
2

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0 7 Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



0 7 . 1 Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer

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0 7 . 2 Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

07 . 3

Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

07 . 4

Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

Do not write outside the box

Extra space

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END OF QUESTIONS